A Course Based Project Report on

CUSTOMER QUEUE SYSTEM

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CSE-Data Science

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CERTIFICATE

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DECLARATION

We declare that the course based project work entitled "**Customer Queue System**" submitted in the Department of **CSE-(CyS, DS) and AI&DS**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in CSE-Data Science** is a bonafide record of our own work carried out under the supervision of **Mrs. N. Madhuri, Dr. G. LN Jayaprada , Department of CSE-(CyS, DS) and AI&DS, VNRVJIET.** Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

Queue management is the systematic control and optimisation of client queues in service- oriented workplaces in order to improve efficiency, consumer satisfaction, and overall operational performance. Customer lineups are unavoidable in businesses such as retail stores, banks, healthcare institutions, and service centres, and handling them properly is critical for creating a great customer experience. Queue management's major purpose is to reduce wait times, eliminate congestion, and ensure a smooth flow of clients through the service process.

Businesses may enhance not just client satisfaction but also operational efficiency and employee productivity by employing efficient queue management. Reduced wait times result in a more positive opinion of the service, which fosters customer loyalty and positive word-of-mouth. Finally, queue management is a strategic strategy that balances customer service aspirations with practical realities, resulting in a win-win situation for both enterprises and their customers in a variety of service-oriented sectors.

TABLE OF CONTENTS

SNo	Title	Page No.
1.	Introduction	1
2.	Existing Solution	2-3
3.	Proposed Solution	4
4.	Algorithms	5
5.	Implementation	6-13
6.	Conclusion	14
7.	Future Enhancements	15
8.	References	16

INTRODUCTION

Queue management is the systematic control and optimisation of client queues in service-oriented workplaces in order to improve efficiency, consumer satisfaction, and overall operational performance. our system integrates multiple applications that dynamically allot seats depending on individual consumer desires, with the goal of prioritising client preferences and improving operational efficiency. Customers are invited to indicate their priority level as a primary input, allowing for a personalised and adapted queuing experience. This novel method not only alleviates the difficulties associated with traditional first-come, first-served models, but also allows businesses to respond to the specific wants and expectations of their customers.

Our system stands out by utilizing a sophisticated algorithmic framework called priority framework. This predictive intelligence enables businesses to strategically allocate resources, optimizing seat assignments and reducing overall wait times. By prioritizing individual preferences, the allocation of seats becomes a dynamic and responsive process, ensuring that loyal customers, special-needs individuals, or those with specific requirements are accorded the attention.

EXISTING SYSTEMS

Physical Queue Management System : A physical queue management system is a manual approach to organizing and controlling customer lines. It involves the use of physical barriers, signage to guide customers in an orderly fashion.

Concepts and technologies used in physical queue management are:

- **Token Systems**: Customers receive a numbered token upon arrival, and they are served in numerical order. This system is often used in banks, government offices, and service centers.
- **Linear Queues**: Customers form a single-file line and are served on a first-come, first-served basis. This is a simple and common approach used in various settings.
- **Multi-Line Queues**: Multiple service counters each have their own queue, and customers choose the shortest line. This is common in supermarkets and retail stores.
- **Appointment Systems**: Customers schedule appointments in advance, reducing the need to wait in a physical queue.

Advantages of existing queue mangement systems:

- **Enhanced Customer Satisfaction:**
  The system allows customers to set their priority levels, leading to a more personalized and customer-centric queuing experience.

- **Operational Efficiency:**
  Predictive analytics and real-time monitoring enable businesses to anticipate and adapt to fluctuations in customer traffic, optimizing resource allocation and staff planning.

- **Personalized Service**:
  The system's ability to consider individual customer profiles and preferences allows for personalized service delivery.

- **Feedback Mechanism:**
  Integration of a feedback mechanism enables businesses to gather insights into customer experiences.

Drawbacks of existing queue management System:

- **Complex Implementation:**
  The deployment of a sophisticated queue management system may involve a complex implementation process, requiring significant time and resources for integration with existing systems and staff training.

- **Limited Customization:**
  Some off-the-shelf queue management systems may have limitations in terms of customization. Businesses with unique or complex workflows may find it challenging to tailor the system to their specific needs.

- **Long wait hours:**
  Few of existing systems has long wait drawback. In order to get served customers are required to wait for longer time.

- **Scalability Issues:**
  As businesses grow, the demand for scalable queue management systems increases. Some existing systems may struggle to efficiently handle a growing volume of customers and transactions without significant upgrades.

- **Overemphasis on Technology:**
  Relying solely on technology without considering the human aspect of customer service can result in a lack of personalization.

PROPOSED SOLUTION

Our proposed Customer Queue Management solution is a system that uses cutting-edge technology to revolutionise seat allocation. This cutting-edge system includes a number of applications that dynamically allot seats depending on consumer priority, offering a personalised and efficient queuing experience. At the core of our solution is a robust algorithm that leverages customer-provided priority inputs. When a customer initiates the queue, they are prompted to specify their priority levels. This priority information becomes instrumental in shaping the allocation strategy, allowing the system to intelligently serve in accordance with individual preferences and importance.

Advantages of Customer Queue Management:

1. **Personalized Service**: By prompting customers to specify their priority levels, the system tailors seat allocations to individual preferences. This personalization enhances customer satisfaction as they receive a service that aligns with their unique needs and expectations.

2. **Optimized Resource Allocation**: The priority information gathered during the initiation of the queue serves as a crucial factor in shaping the allocation strategy. High-priority customers receive preferential treatment, ensuring that critical resources are allocated where they are most needed.

3. **Reduced Wait Times**: This allocation strategy, driven by customer-provided priority inputs, facilitates quicker and more efficient seat assignments. High-priority customers are expedited through the queue, resulting in reduced wait times and an overall improvement in the customer experience.

4. **Enhanced Customer Satisfaction:** The system's ability to serve customers in accordance with their individual preferences contributes to a more positive and satisfactory experience.

Customer Queue Management system, centered around a robust algorithm leveraging customer-provided priority inputs, extend beyond mere operational efficiency. The system's ability to provide personalized service, optimize resource allocation, reduce wait times.

ALGORITHM

.

1. **Initialize and Input System Information**:
    - Prompt the user to enter the system name (sysName).
    - Prompt the user to enter the customer name (custName).
2. **Input Number of Processes**:
    - Prompt the user to enter the number of processes (n).
3. **Process Input Details**:
    - For each process (from 1 to n):
        - Assign process_id to be the loop index (1-based).
        - Prompt the user to enter the service_type based on system name:
            - For "hospital": EmergencyCare, ScheduledAppointments, Diagnostics, Pharmacy.
            - For "restaurant": DineIn, Takeaway, Delivery.
        - Prompt the user to enter the priority:
            - For both systems: Critical, High, Medium, Low.
        - Calculate the sorting_value for each process using calculateSortingValue(service_type, priority).
4. **Sort Processes**:
    - Use mergeSort to sort the processes in descending order based on sorting_value. In case of ties, use process_id to break the tie.
5. **Display Sorted Process Execution**:
    - Print the system name and the sequence of process execution based on the sorted order.

IMPLEMENTATION

CODE:

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

// Define enum for type of service
enum class ServiceType {
    EmergencyCare,
    ScheduledAppointments,
    Diagnostics,
    Pharmacy,
    DineIn,
    Takeaway,
    Delivery
};

// Define enum for customer priority
enum class CustomerPriority {
    Critical,
    High,
    Medium,
    Low,
    VIPRegular,
    ReservationHolder,
    WalkIn,
    OnlineOrder
};

struct Process {
    int process_id;
    ServiceType service_type;  // Type of service for the process
    CustomerPriority priority; // Customer priority for the process
    int sorting_value;         // Value used for sorting
};

// Function to calculate sorting value based on service type and priority

int calculateSortingValue(ServiceType serviceType, CustomerPriority priority) {
    // Assign scores based on service type
    int serviceScore = 0;
    switch (serviceType) {
        case ServiceType::EmergencyCare:
```

```
        serviceScore = 200;
        break;
    case ServiceType::ScheduledAppointments:
        serviceScore = 100;
        break;
    case ServiceType::Diagnostics:
        serviceScore = 80;
        break;
    case ServiceType::Pharmacy:
        serviceScore = 60;
        break;
    case ServiceType::DineIn:
        serviceScore = 90;
        break;
    case ServiceType::Takeaway:
        serviceScore = 70;
        break;
    case ServiceType::Delivery:
        serviceScore = 50;
        break;
}

// Assign scores based on customer priority
int priorityScore = 0;
switch (priority) {
    case CustomerPriority::Critical:
        priorityScore = 150;
        break;
    case CustomerPriority::High:
        priorityScore = 100;
        break;
    case CustomerPriority::Medium:
        priorityScore = 50;
        break;
    case CustomerPriority::Low:
        priorityScore = 25;
        break;
    case CustomerPriority::VIPRegular:
        priorityScore = 120;
        break;
    case CustomerPriority::ReservationHolder:
        priorityScore = 90;
        break;
    case CustomerPriority::WalkIn:
        priorityScore = 60;
        break;
    case CustomerPriority::OnlineOrder:
```

```
            priorityScore = 30;
            break;
        }

        // Combine service score and priority score with weights
        // Here, giving more weight to service type (e.g., 0.7) than to priority (e.g., 0.3)
        double serviceWeight = 0.7;
        double priorityWeight = 0.3;

        int sortingValue = static_cast<int>(serviceScore * serviceWeight + priorityScore * priorityWeight);

        return sortingValue;
    }


    // Merge function for merge sort (descending order)
    void merge(vector<Process> &processes, int left, int mid, int right) {
        int n1 = mid - left + 1;
        int n2 = right - mid;

        vector<Process> L(n1);
        vector<Process> R(n2);

        for (int i = 0; i < n1; i++)
            L[i] = processes[left + i];
        for (int j = 0; j < n2; j++)
            R[j] = processes[mid + 1 + j];

        int i = 0, j = 0, k = left;
        while (i < n1 && j < n2) {
            // Change comparison for descending order
            if (L[i].sorting_value > R[j].sorting_value ||
                (L[i].sorting_value == R[j].sorting_value && L[i].process_id < R[j].process_id)) {
                processes[k] = L[i];
                i++;
            } else {
                processes[k] = R[j];
                j++;
            }
            k++;
        }

        while (i < n1) {
            processes[k] = L[i];
            i++;
            k++;
        }
```

```
        while (j < n2) {
            processes[k] = R[j];
            j++;
            k++;
        }
    }

    // Merge sort function
    void mergeSort(vector<Process> &processes, int left, int right) {
        if (left >= right) {
            return;
        }
        int mid = left + (right - left) / 2;
        mergeSort(processes, left, mid);
        mergeSort(processes, mid + 1, right);
        merge(processes, left, mid, right);
    }

    void displayOptions(string sysName) {
        if (sysName == "hospital") {
            cout << "Types of Services:\n";
            cout << "1. Emergency Care\n";
            cout << "2. Scheduled Appointments\n";
            cout << "3. Diagnostics\n";
            cout << "4. Pharmacy Services\n";
            cout << "Customer Priorities:\n";
            cout << "1. Critical\n";
            cout << "2. High\n";
            cout << "3. Medium\n";
            cout << "4. Low\n";
        } else if (sysName == "restaurant") {
            cout << "Types of Services:\n";
            cout << "1. Dine-In\n";
            cout << "2. Takeaway\n";
            cout << "3. Delivery\n";
            cout << "Customer Priorities:\n";
            cout << "1. VIP/Regular Customers\n";
            cout << "2. Reservation Holders\n";
            cout << "3. Walk-Ins\n";
            cout << "4. Online Orders\n";
        }
    }
void customerQueueManagement(string sysName, string custName) {
int n;
        cout << "Enter the number of processes: ";
```

```
    cin >> n;

    vector<Process> processes(n);

    // Display options based on system type
    displayOptions(sysName);

    // Input process details for
    (int i = 0; i < n; i++) {
        cout << "Process-" << i + 1 << ":\n"; processes[i].process_id
        = i + 1;

        int serviceChoice, priorityChoice;
        cout << "Enter type of service (1-"; if
        (sysName == "hospital")
            cout << "4): ";
        else if (sysName == "restaurant") cout
            << "3): ";
        cin >> serviceChoice;

        // Set service type based on user input
        switch (serviceChoice) {
            case 1:
                processes[i].service_type = ServiceType::EmergencyCare;
                break;
            case 2:
                processes[i].service_type = ServiceType::ScheduledAppointments; break;
            case 3:
                processes[i].service_type = ServiceType::Diagnostics;
                break;
            case 4:
                if (sysName == "hospital")
                    processes[i].service_type = ServiceType::Pharmacy;
                else if (sysName == "restaurant")
                    processes[i].service_type = ServiceType::Delivery;
                break;
            default:
                processes[i].service_type = ServiceType::EmergencyCare; // Default to EmergencyCare
                break;
        }

        cout << "Enter customer priority (1-4): ";
        cin >> priorityChoice;
// Set customer priority based on user input
switch (priorityChoice) {
```

```
            case 1:
                processes[i].priority = CustomerPriority::Critical;
                break;
            case 2:
                processes[i].priority = CustomerPriority::High;
                break;
            case 3:
                processes[i].priority = CustomerPriority::Medium;
                break;
            case 4:
                processes[i].priority = CustomerPriority::Low;
                break;
            default:
                processes[i].priority = CustomerPriority::Low; // Default to Low priority
                break;
        }

        // Calculate sorting value based on service type and priority
        processes[i].sorting_value = calculateSortingValue(processes[i].service_type, processes[i].priority);
    }

    // Sort processes using merge sort based on sorting_value and process_id
    mergeSort(processes, 0, n - 1);

    cout << "\t" << sysName << " Queue Management\t\n";
    cout << "Sequence of execution : ";
    for (int i = 0; i < n; i++) {
        if (i != n - 1)
            cout << custName << "-" << processes[i].process_id << " --> ";
        else
            cout << custName << "-" << processes[i].process_id;
    }
}

int main() {
    string sysName, custName;
    cout << "Enter System Name (hospital/restaurant):\n";
    cin >> sysName;
    cout << "Enter customer name\n";
    cin >> custName;
    customerQueueManagement(sysName, custName);
    return 0;
}
```

OUTPUT

OUTPUT 1: Hospital Queue Management System

```
Enter System Name (hospital/restaurant):
hospital
Enter customer name
patient
Enter the number of processes: 3
Types of Services:
1. Emergency Care
2. Scheduled Appointments
3. Diagnostics
4. Pharmacy Services
Customer Priorities:
1. Critical
2. High
3. Medium
4. Low
Process-1:
Enter type of service (1-4): 1
Enter customer priority (1-4): 3
Process-2:
Enter type of service (1-4): 2
Enter customer priority (1-4): 2
Process-3:
Enter type of service (1-4): 4
Enter customer priority (1-4): 1
        hospital Queue Management
Sequence of execution : patient-1 --> patient-2 --> patient-3

...Program finished with exit code 0
Press ENTER to exit console.
```

OUTPUT 2: Restaurant Queue Management System

```
Enter System Name (hospital/restaurant):
restaurant
Enter customer name
customer
Enter the number of processes: 4
Types of Services:
1. Dine-In
2. Takeaway
3. Delivery
Customer Priorities:
1. VIP/Regular Customers
2. Reservation Holders
3. Walk-Ins
4. Online Orders
Process-1:
Enter type of service (1-3): 1
Enter customer priority (1-4): 3
Process-2:
Enter type of service (1-3): 2
Enter customer priority (1-4): 1
Process-3:
Enter type of service (1-3): 3
Enter customer priority (1-4): 4
Process-4:
Enter type of service (1-3): 3
Enter customer priority (1-4): 3
        restaurant Queue Management
Sequence of execution : customer-1 --> customer-2 --> customer-4 --> customer-3

...Program finished with exit code 0
Press ENTER to exit console.
```

FUTURE ENHANCEMENTS

The algorithm for the customer queue management system presented here can be expanded and enhanced in various ways to make it more robust, flexible, and applicable to a wider range of scenarios. Here are some potential future enhancements:

1. **Dynamic Priority Adjustment**:
    - Implement dynamic priority adjustments based on waiting time. For instance, if a process has been waiting for too long, its priority could be incremented to avoid starvation.
2. **Additional Service Types and Priorities**:
    - Expand the range of service types and priorities to cover more specific scenarios within hospitals and restaurants, or to make the system applicable to other industries like banks, customer support centers, etc.
3. **Real-time Data Integration**:
    - Integrate real-time data feeds to adjust priorities dynamically based on current conditions, such as real-time updates of emergency cases in hospitals or peak hours in restaurants.
4. **Machine Learning for Priority Prediction**:
    - Use machine learning to predict and adjust priorities based on historical data and patterns, making the system smarter and more proactive in managing queues.

CONCLUSION

In conclusion, this customer queue management project, integrated within the operating system framework, represents a significant leap forward in redefining the landscape of customer service efficiency. Through a meticulous literature survey and the implementation of cutting-edge technologies, the project has successfully addressed critical challenges in traditional queue management systems. The seamless integration of artificial intelligence, real-time analytics, and user-centric mobile applications has not only streamlined queue processes but has also empowered customers with greater control over their waiting experiences. The dynamic adaptation of queue priorities based on customer profiles and real-time factors further underscores the project's commitment to providing personalized and responsive service.

The achievements of the project, including reduced waiting times, optimized resource allocation, and improved customer satisfaction, highlight the tangible benefits of leveraging operating systems for enhanced queue management. This ensures a holistic approach to customer interactions, fostering deeper connections and understanding.

Looking forward, scalability, continuous improvement through data-driven insights, and a commitment to security and privacy considerations stand out as critical elements for the sustained success of such systems. The project sets the stage for future endeavors, urging a balance between automation and human touch, and emphasizing the ethical implications of handling sensitive customer data.

In essence, this project not only marks a technological milestone but also signifies a paradigm shift towards a more responsive, adaptive, and customer-focused approach to queue management, with implications that extend beyond the realms of the current project scope.

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